

Tawaun A. Lucas, Ph.D.

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Staff-level computational biologist with a Ph.D. from Stanford and 8+ years of experience developing and validating quantitative methods on high-dimensional genomic data across academic and industry settings. Deep expertise in NGS analysis, statistical algorithm development, and scalable HPC and cloud workflows, with a track record of translating complex multi-omic datasets into defensible scientific claims that drive product and clinical decisions. Proven technical leader with experience mentoring junior scientists and acting as subject matter expert across cross-functional teams.

Experience

Senior Computational Biologist, Spatial Omics

LatchBio

San Francisco, CA

Mar. 2026 – Present

- **Algorithm Benchmarking:** Translate end-to-end spatial transcriptomics analysis workflows into structured, reproducible patterns with defined ground truth and known failure modes, enabling rigorous evaluation of AI agent outputs against defensible scientific standards.
- **Agent Evaluation:** Review and grade AI-generated analyses across sequencing-based and image-based spatial platforms (Xenium, Visium, MERFISH), Multiome platforms and Perturb-seq applying independent scientific judgment to distinguish outputs that look correct from those that are demonstrably correct.
- **Production Pipeline Development:** Develop and maintain production-grade single-cell and spatial analysis workflows used to train and validate biological reasoning agents across diverse assay types.
- **Cross-Functional Technical Leadership:** Partner directly with CTO and head of product to define new approaches for automating high-consequence analytical decisions in preclinical R&D.

Senior Computational Biologist

Chan Zuckerberg Biohub

Redwood City, CA

Nov. 2025 – April 2026

- **Platform Architecture:** Led development of the [AI Workspace](#), a scalable cloud platform automating scRNA-seq analysis workflows and standardizing outputs for rigorous downstream interpretation.
- **Quantitative Validation:** Designed and executed benchmarking studies for AI/ML foundation models, establishing evaluation criteria to ensure scientific rigor in target identification from high-dimensional single-cell data.
- **Open-Source Tool Development:** Partnered with software engineers and biologists to build and deploy open-source pipelines enabling the broader research community to analyze multimodal spatial and single-cell datasets.

Computational Biologist (Contract)

Genentech Inc.

South San Francisco, CA

Aug. 2025 – Present

- **Target Identification:** Served as bioinformatics technical lead for IBD immunology program, integrating transcriptomic, histological, and clinical data to prioritize two therapeutic targets for advancement.
- **Algorithm Development:** Developed statistical and ML-based computational frameworks for multi-modal data integration, generating quantitative hypotheses that directly informed R&D decision-making.
- **Process Improvement:** Engineered automated analysis workstreams that reduced dataset turnaround time by approximately 10%, accelerating experimental cycles and enabling faster go/no-go decisions.

Postdoctoral Fellow

Genentech, Inc.

South San Francisco, CA

April 2021 – August 2025

- **Large-Scale NGS Analysis:** Constructed a cross-disease single-cell transcriptomic atlas of human and mouse astrocytes, developing and applying custom statistical methods to characterize cell-state heterogeneity across neurodegeneration indications.
- **Spatial Transcriptomics Pipelines:** Led end-to-end development of analysis pipelines for spatial transcriptomics platforms (10x Visium, Xenium), integrating ML algorithms to jointly model gene expression and spatial architecture.
- **Multi-Omic Algorithm Development:** Developed quantitative frameworks unifying Perturb-seq, CITE-seq, and scATAC-seq to identify novel inflammatory targets, providing mechanistic rationale for each nominated candidate.
- **Technical Mentorship:** Trained and guided 3 junior researchers, 1 research associate, and 2 interns; served as subject matter expert in cross-functional neuroscience and immunology project teams.

- **Reproducible Pipeline Development:** Built and deployed Snakemake-based RNA-seq analysis workflows on HPC environments, establishing reproducible methods for neuroimmune transcriptomic studies.
- **Statistical Analysis:** Applied differential expression, pathway analysis, and dimensionality reduction methods to characterize a novel peripheral glial subtype, generating findings published in peer-reviewed literature.
- **Tool Development:** Developed interactive R Shiny applications for real-time visualization of transcriptomic data, adopted by multiple academic labs for self-serve exploratory analysis.

Education

Stanford University School of Medicine
Ph.D. in Neurosciences

Sept. 2014 – May 2021

California State University, Northridge
B.A. in Psychology (Honors)

Aug. 2009 – May 2014

Technical Skills

Domain Expertise: NGS Analysis, Diagnostic Assay Development, Biomarker Discovery, Target Identification, Statistical Algorithm Development, Multi-Omic Integration

Genomics & Assays: scRNA-seq, snRNA-seq, Spatial Transcriptomics (Visium, Xenium, MERFISH), scATAC-seq, CITE-seq, Perturb-seq, 10x Multiome, Bulk RNA-seq, Cell Atlases

Statistical & ML Methods: Differential Expression, Dimensionality Reduction, Supervised/Unsupervised ML, Deep Learning (PyTorch, TensorFlow), Pathway Analysis, Bayesian Methods

Languages & Infrastructure: Python (Pandas, NumPy, Scikit-learn, Scanpy), R (Tidyverse, Shiny, Seurat), Snakemake, Argo Workflows, AWS, HPC/SLURM, Git, Docker

Selected Publications

SpatialAgent: An Autonomous AI Agent for Spatial Biology

March 2025

Wang, H. ..., Lucas, T A ..., Regev, A. *Submitted*. [Preprint: doi:10.1101/2025.04.03.646459v1](https://doi.org/10.1101/2025.04.03.646459v1)

Integrated Cross-Disease Atlas of Human and Mouse Astrocytes Reveals Heterogeneity...

Feb. 2025

Lucas, T A, Novikova, G. ..., Kaminker, J. *In Revision – Nature Neuroscience*. [Preprint: doi:10.1101/2025.02.12.637903](https://doi.org/10.1101/2025.02.12.637903)

Translatome analysis reveals microglia and astrocytes to be distinct regulators of inflammation...

Aug. 2023

Hernandez, V. ..., Lucas, T A ..., Buckwalter, M. *Glia*. [doi:10.1002/glia.24377](https://doi.org/10.1002/glia.24377)

Spleen glia are a transcriptionally unique glial subtype interposed between immune cells...

March 2021

Lucas, T A, Zhu, L., Buckwalter, M. *Glia*. [doi:10.1002/glia.23993](https://doi.org/10.1002/glia.23993)

Leadership & Service

- **Reviewer:** Nature Neuroscience, Journal of Clinical Investigation, Glia
- **Founder:** Diversity Center of Representation and Empowerment (DCORE) at Stanford Medicine
- **Fellowships:** NSF Graduate Research Fellowship (GRFP), DARE Fellowship, NIH MARC Fellowship
- **Awards:** Stanford Biosciences Excellence in Diversity; Commencement Speaker (2020)